MILITARY SPECIFICATION

AMPLIFIER, COMPASS, SERVO (TYPE MF-1 COMPASS)

This amendment forms a part of Military Specification MIL-A-22165(Aer) of 15 August 1959, and has been approved by the Bureau of Naval Weapons, Department of the Navy.

By this amendment basic MIL-A-22165(Aer) is changed to read MIL-A-22165(Wep).

Page 3: Paragraph 3.5 Design and Construction.- Revise third sentence to read:

"The cam compensator shall have visual compensated and uncompensated indicators and a vernier calibrated in 0.1 degree".

Page 4: Paragraph 3.12 Dial Marking.- Add the following sentence and table to the end of this paragraph:

"The dimensions of the dial markings shall be as follows:

	Compensated Dial		Uncompensated Dial	
	Height or Length Inch	Wide Inch ±.002	Height or Length Inch	Width Inch ±.002
Numerals, Letters	.070	.012	.070	.012
10° Graduations	.125	.010	.094	.010
5° Graduations	.125	.006	.094	.006
1° Graduations	.078	.006	.062	.006

	Vernier	
	Height or Length Inch	Wide Inch ± .002
Numerals	.062	.012
"0" and "1" Graduations	.060	.010
6th Graduation	.060	.006
All others	.046	.006

Page 5: Delete Figure 1 and substitute new Figure 1 attached.

Page 7: Add paragraph 4.3.2:

Authorization.- Requests for authorization for Qualification Tests shall be addressed to the Bureau of Naval Weapons, Department of the Navy, Washington 25, D. C., the qualifying agency.

FSC 6605

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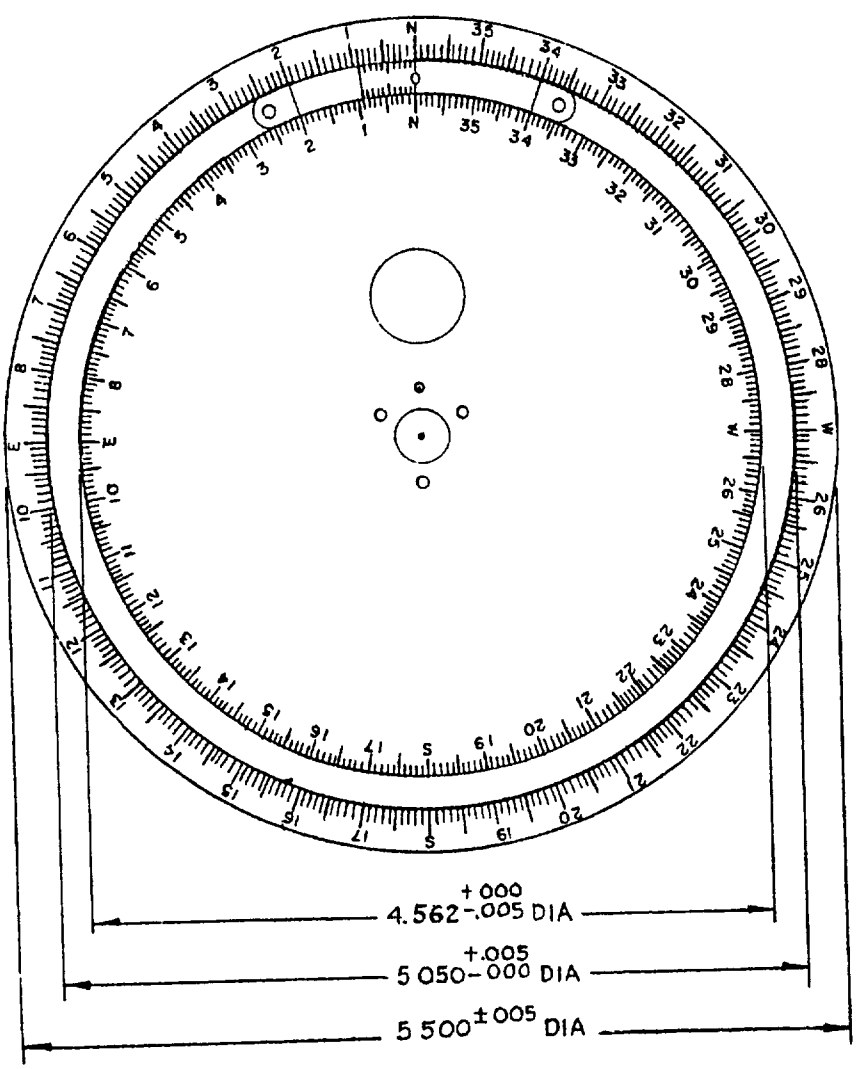


Figure 1

Compensated and Uncompensated Dials and Vernier

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Page 10: Paragraph 4.7.2 Continuity. - Under Resistance (Ohms) opposite Pin Designations "J2004-29 to J2004-30 (Type II Only)" and "J2004-30 to J2004-31 (Type II Only)" change the value "4.5± 1" to "45 ± 10" in two places.

Under Resistance (Ohms) opposite Pin Designations "J2004-32 to J2004-33 (Type II Only)" change the value "18.8 ± 4" to "188 ± 40".

Page 11: Paragraph 4.7.3 Dielectric Strength.- Add the following as the last sentence:

Insulation breakdown shall be defined as an alternating current leakage impedance of less than 200,000 ohms, or a leakage current of 2.5 milliamperes.

Page 12: Paragraph 4.7.5.3 Latitude Correction Voltage Regulation.- In the first and last sentences, change the voltage value from "9.0 volts dc" to "9.5 volts dc".

Page 14: Paragraph 4.7.6.7.1.1 (Type I Only) Section c.- Change second sentence to read:

"Each successive null voltage reading between J2004-21 to J2004-22 shall not change more than 10 mv from the preceding null voltage reading."

Page 14: Paragraph 4.7.6.7.1.2 (Type II Only) Section (c).- Change second sentence to read:

"Each successive null voltage reading between J2004-35 to J2004-34 shall not change more than 50 mv from the preceding null voltage reading."

Page 15: Paragraph 4.7.6.7.4.2 (Type II Only).- In section (a) change the voltage "600 millivolts" to "5 volts".

Page 15: Paragraph 4.7.6.7.5.2 (Type II Only).- In section (a) change the voltage "600 millivolts" to "5 volts".

Page 15: Paragraph 4.7.6.7.6.1 (Type I Only).- In section (c) revise last sentence to read:

"The voltage from J2004-21 to J2004-22 shall not be greater than 10 millivolts."

Page 17: Paragraph 4.7.9.1 Fast Synchronization.- In both sections (a) and (b) change the figures "5.6 ±1.4" to "6.3 ± 2.1".

Page 17: Paragraph 4.7.13 High Temperature Operation.- Revise fourth sentence to read:

"At the end of the 4 hour period and with the temperature maintained at 71° C (160°F), the amplifier shall be subjected to and meet the requirements of the following sections.

(a) Section 4.7.6.4

(b) Section 4.7.6.5

(c) Section 4.7.6.6

(d) Section 4.7.6.7.1 Null, except as follows:

Section 4.7.6.7.1.1 (Type I Only) - The null voltages in each part shall be (a) 20 mv, (b) 25 mv, and (c) 40 mv.

Section 4.7.6.7.1.2 (Type II Only) - The null voltages in each part shall be (a) 80 mv, (b) 90 mv, and (c) 160 mv.

(e) Section 4.7.6.7.2 Sensitivity, Phase and Limits, except as follows:

Section 4.7.6.7.2.1 (Type I Only) - The voltage in part (a) shall be .660 ± .13 at a phase of 7 ± 4 degrees.

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Section 4.7.6.7.2.2 (Type II Only) - The voltage in part (a) shall be 5.9 ± .9 at a phase of 20 ± 6 degrees.

(f) Section 4.7.6.7.3, except as follows:

Section 4.7.6.7.3.1 (Type I Only) - The voltage in part (a) shall be .660 ± .13 at a phase of -173 ± 4 degrees.

Section 4.7.6.7.3.2 (Type II Only) - The voltage in part (a) shall be 5.9 ± .9 at a phase of -200 ± 6 degrees.

(g) Section 4.7.8.1

(h) Section 4.7.8.2.1 - (Except in part (c) the servo repeater autosyn dial pointer shall move to zero degrees ± 1.0 degree.

(i) Section 4.7.9.2 - (Except in parts (a) and (b) the repeater autosyn dial pointer shall turn through 180 degrees ± 70 degrees.

(j) Section 4.7.10"

Page 17: Paragraph 4.7.14 Low Temperature Operation.- Revise fifth sentence to read:

"At the end of the 4 hour period and with the temperature maintained at -54° C (-65°F), the amplifier shall be subjected to and meet the requirements of the following sections:

(a) Section 4.7.6.4

(b) Section 4.7.6.5

(c) Section 4.7.6.6

(d) Section 4.7.6.7.1 Null, except as follows:

Section 4.7.6.7.1.1 (Type I Only) - The null voltages in each part shall be (a) 20 mv, (b) 25 mv, and (c) 40 mv.

Section 4.7.6.7.1.2 (Type II Only) - The null voltages in each part shall be (a) 80 mv, (b) 90 mv, and (c) 160 mv.

(e) Section 4.7.6.7.2 Sensitivity, Phase and Limits, except as follows:

Section 4.7.6.7.2.1 (Type I Only) - The voltage in part (a) shall be .660 ± .13 at a phase of 7 ± 4 degrees.

Section 4.7.6.7.2.2 (Type II Only) - The voltage in part (a) shall be 5.9 ± .9 at a phase of 20 ± 6 degrees.

(f) Section 4.7.6.7.3, except as follows:

Section 4.7.6.7.3.1 (Type I Only) - The voltage in part (a) shall be .660 ± .13 at a phase of -173 ± 4 degrees.

Section 4.7.6.7.3.2 (Type II Only) - The voltage in part (a) shall be 5.9 ± .9 at a phase of -200 ± 6 degrees.

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(g) Section 4.7.8.1

(h) Section 4.7.8.2.1 - (Except in part (c) the servo repeater autosyn dial pointer shall move to zero degrees ± 1.0 degree).

(i) Section 4.7.9.2 - (Except in parts (a) and (b) the repeater autosyn dial pointer shall turn through 180 degrees ± 70 degrees).

(j) Section 4.7.10"

Page 18: Paragraph 4.7.15 High Temperature Exposure.- Revise sixth sentence to read the same as the change for Paragraph 4.7.13 (above).

Page 18: Paragraph 4.7.16 Low Temperature Exposure.- Revise third sentence to read the same as the change for Paragraph 4.7.14 (above).

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